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1. Author and Contact Information

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 - **Editor** of VDMA OPC UA Industrial Joining Technologies Working Group.
- Contact for any questions/updates/support on using the demo and extending it.

2. Prerequisites

- **Windows Binary**
 - Windows 10 or later (Built using Windows SDK Version: **10.0.26100**).
 - Download **Visual Studio Redistributable**: [VC-Redist Download](#)
- **Docker Image**: Ensure that Docker is installed and running.
- **OPC UA Test Client**: Download and install any OPC UA Client. **Example**: [UaExpert Download](#)
- **IJT CS Reference**:
 - [OPC 40450 Joining Base](#)
 - [OPC 40451 Tightening](#)

3. Installation of Application Files

- Download the following files in the **Installation Directory: OPC UA IJT Server Simulator**.
 - **opcua_ijt_demo_application.exe**
 - Contains **multiple** NodeSet files in XML format as below:
 - Opc.Ua.XXX.NodeSet2.xml
 - **Optional Files**
 - **Multiple JSON files for simulation**
 - server_configuration.json
 - simulated_asset_data.json
 - ...

4. Running the application

- **Common Steps**
 - Go to the “**OPC_UA_IJT_Server_Simulator**” directory.
 - The **EndpointUrl** of the OPC UA Server would be:
 - **opc.tcp://localhost:40451** or **opc.tcp://YourComputerName:40451**.
- **Windows Binary**
 - Ensure that the user has **Read/Write access** to the **Installation Directory**.
 - **Launch** the binary file (**opcua_ijt_demo_application.exe**).
 - Run as Administrator or at least with **Read/Write** access.
- **Docker Image**
 - Navigate to the **Release2** directory (one level above **OPC_UA_IJT_Server_Simulator**).
 - Run the following command:
 - `docker compose up`
 - Or manually:
 - `docker build -t opcua-ijt-server .`
 - `docker run --rm -p 40451:40451 opcua-ijt-server`
 - To **connect** from a remote machine, pass the host IP:
 - `docker run --rm -p 40451:40451 -e OPCUA_HOSTNAME=192.168.1.10 opcua-ijt-server`

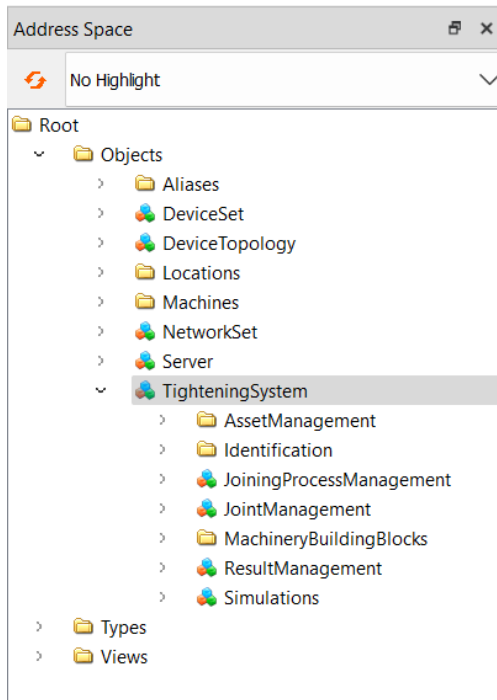
5. Access data from OPC UA Client

- Launch the OPC UA Client and connect to the given **EndpointUrl**.
- It will show the primary entry point: **TighteningSystem**.
- **All** the **Nodes** shown below are as per the **Companion Specification**.
- The **Simulations** node is the Application Node.

6. Validate if the Client Supports Required OPC UA Features

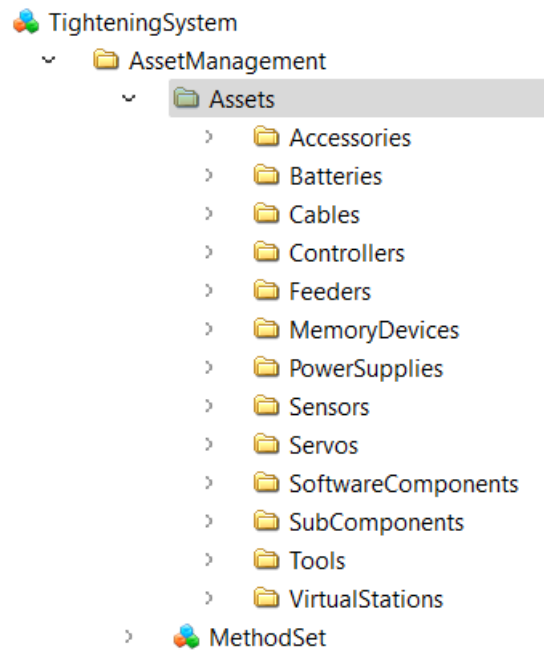
- OPC UA Client **shall support** OPC UA **Extension** Objects (Custom Structure Types) to **consume** the data as per the OPC UA IJT Standard.
- The quick test to validate if the Client application supports **extension** objects is to subscribe to **Result variable** or **Result Event** and visualize if the data is readable from the Client.
- Refer to **upcoming** sections on how to **subscribe** to **Result** variable or **Result** Event.

7. Tightening System Entry Point - AddressSpace View



8. Asset Data

Browse the respective Asset Nodes from the address space and subscribe/read the respective data. **Drag** and **Drop** the **variables** to subscribe in the **Data Access View** window.



Example Asset Address Space View

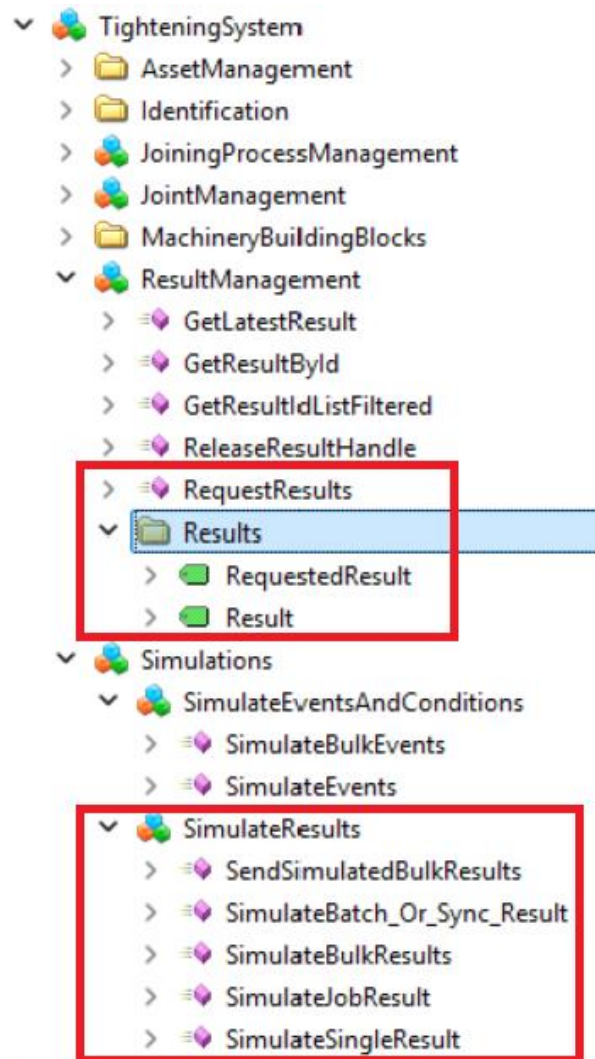
- ▼  TighteningSystem
 - ▼  AssetManagement
 - ▼  Assets
 - >  Accessories
 - >  Batteries
 - >  Cables
 - >  Controllers
 - >  Feeders
 - >  MemoryDevices
 - >  PowerSupplies
 - >  Sensors
 - >  Servos
 - >  SoftwareComponents
 - >  SubComponents
 - ▼  Tools
 - ▼  TighteningTool
 - >  Identification
 - >  LifetimeCounters
 - >  MachineryBuildingBlocks
 - >  Maintenance
 - >  Monitoring
 - >  OperationCounters
 - >  Parameters
 - >  VirtualStations
 - >  MethodSet

9. Result Data

- **Result Access Options:**
 - Subscribe to the **Result** or **RequestedResult** variable shown below. **Drag** and **Drop** the **variables** to subscribe in the **Data Access View** window.
 - **Result** Variable is intended for getting **live** results.
 - **RequestedResult** Variable is intended for getting **historical** results.
 - Subscribe to **events** by subscribing to the **Server** node in the Event View.
- **Simulation Options**
 - Use the following **three methods** to simulate different types of **Results**. A new **Result** is generated upon the execution of the following **methods**.
 - **SendSimulatedBulkResults**
 - **To send already generated bulk results without recreating.**
 - **SimulateBatch_or_Sync_Result**
 - To generate Batch Result.
 - **SimulateBulkResults**
 - To generate several live Results.
 - **SimulateJobResult**
 - To generate a Job Result.
 - **SimulateSingleResult**
 - To generate a single live Result.
 - The **simulated** data is similar to the examples **defined** in the **Annexure** sections of the Companion Specification.

10. Result, RequestedResult Variable and Result Simulation Methods

Subscribe to Result variable as shown below.



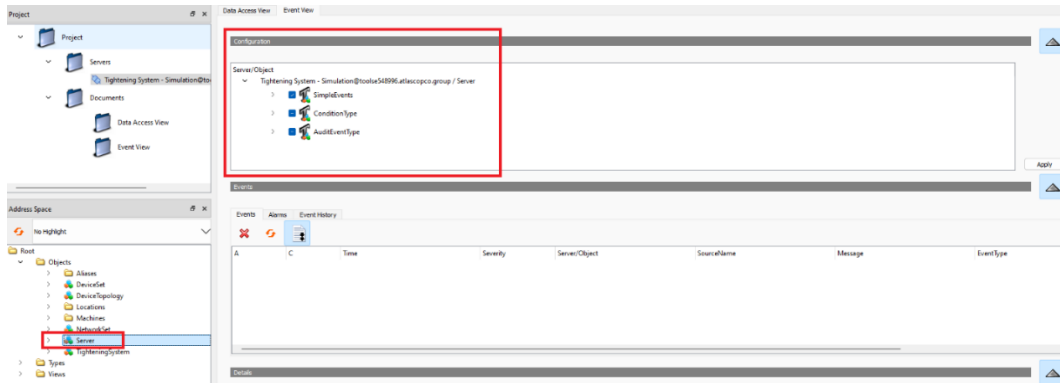
11. Result Data Access View – Subscribed Result Variable Data Example

The outcome of the **Result variable subscription** is shown below.

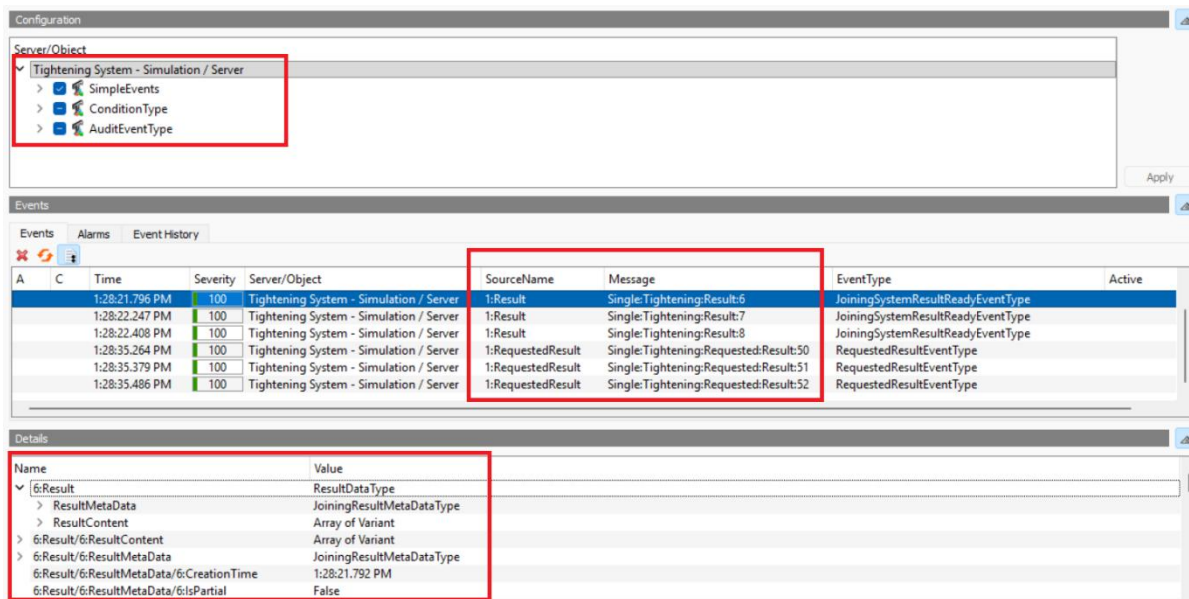
#	Display Name	Value	Datatype	Source Timestamp
1	Result	Double click to display value	ExtensionObject	8:35:14.750 PM
2	ResultContent	Double click to display value	Variant	8:35:14.752 PM
3	ResultMetaData	Double click to display value	ExtensionObject	8:35:14.753 PM
4	AssemblyType	1	Byte	8:35:14.753 PM
5	AssociatedEntities	Double click to display value	ExtensionObject	8:35:14.753 PM
6	Classification	1	Byte	8:35:14.753 PM
7	CreationTime	2025-10-09T18:35:04.654Z	DateTime	8:35:14.753 PM
8	Description	"en", "Single:Tightening:Result:2"	LocalizedText	8:35:14.753 PM
9	ExtendedMetaData	Double click to display value	ExtensionObject	8:35:14.753 PM
10	InterventionType	0	Byte	8:35:14.753 PM
11	IsGeneratedOffline	false	Boolean	8:35:14.753 PM
12	IsPartial	false	Boolean	8:35:14.753 PM
13	IsSimulated	true	Boolean	8:35:14.753 PM
14	JoiningTechnology	"en", "Tightening"	LocalizedText	8:35:14.753 PM
15	Name	Single:Tightening:Result:2	String	8:35:14.753 PM
16	OperationMode	2	Byte	8:35:14.753 PM
17	ProcessingTimes	Double click to display value	ExtensionObject	8:35:14.753 PM
18	ResultCounters	Double click to display value	ExtensionObject	8:35:14.753 PM
19	ResultEvaluation	1 (OK)	Int32	8:35:14.753 PM
20	ResultEvaluationCode	0	Int64	8:35:14.753 PM
21	ResultEvaluationDetails	"en", "OK TIGHTENING"	LocalizedText	8:35:14.753 PM
22	ResultId	15EC5500-CBA9-F648-A323-30BA63754435	String	8:35:14.753 PM
23	ResultState	1	Int32	8:35:14.753 PM
24	SequenceNumber	2	UInt64	8:35:14.753 PM

12. Subscribe to Result Events or Other Events

- Connect to the OPC UA Server using UaExpert or any other OPC UA Client.
- Subscribe to “**Server**” Object. In UaExpert, Drag and Drop the “**Server**” Object in the following **Configuration Window** as shown below.



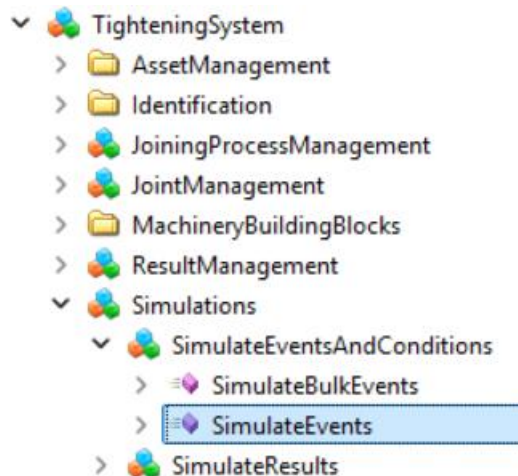
- Select the “**Simple Event**” checkbox and it shall **show** as checked: ✓ and Click on **Apply**.
- Generate a **new Result**, and the Result will be listed in the **Events Window** as shown below.



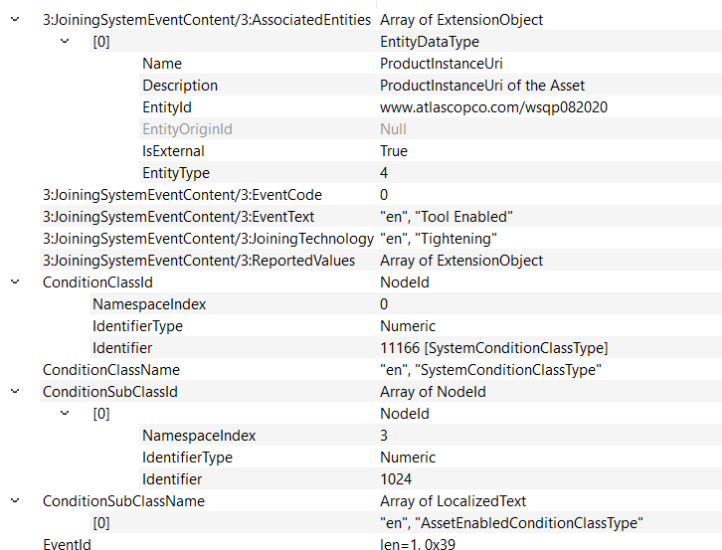
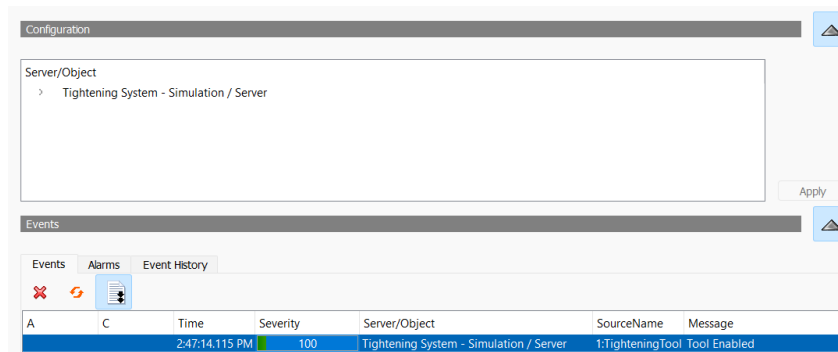
13. Event Data

Invoke the **SimulateEvents** or **SimulateBulkEvents** method as shown below to **generate 50+** different type of events.

Note: The **content** of the Events would be similar to any type of event from a joining system.



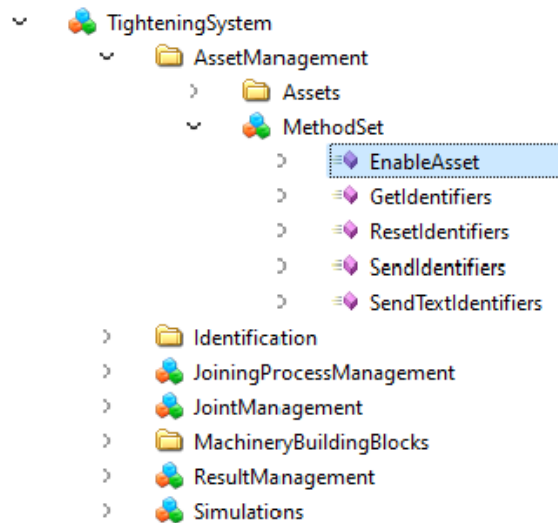
Example Events View



14. Commands Example

An example simulation of **EnableAsset** is provided. It takes the input of the ProductInstanceUri of the Tool.

Few error cases can be simulated when the input argument is empty or invalid. A respective error is shown in the output arguments.



Example Command View

Call EnableAsset on MethodSet

Input Arguments			
Name	Value	DataType	Description
ProductInstanceUri	www.atlascopco.com/wsdp082020	String	
Enable	☒	Boolean	

Output Arguments			
Name	Value	DataType	Description
Status	0	Int64	
StatusMessage	en SUCCESSFUL OPERATION	LocalizedText	

Result

Succeeded

Call Close